Formal languages and automata 2023

Test 4

13 October 2023

Total marks: 25
Time allowance: 45 mins

1. (10 marks)Prove that the formula $\forall x\,(P(x) \wedge Q(x)) \to (\forall x\, P(x) \wedge \forall x\, Q(x))$ is valid (i.e., true in every first-order model).

2. Consider the following signature of a language meant for describing a fixed computation by a fixed Turing machine M (the presumed domain of discourse is natural numbers, which shall be used to number the steps of the computation and the cells of M's tape; it shall be assumed that the tape of M extends infinitely to the right, but has the leftmost cell, numbered 0; we shall need a special symbol, the end-of-tape marker, which shall always be written in the leftmost cell, so that M knows not to move left of the leftmost cell; the states of M are assumed to be q_0, q_1, etc., with q_0 being the starting state, q_1 the accepting state, and q_2 the rejecting state; the symbols of M's alphabet are assumed to be s_0, s_1, etc., with s_0 being the blank and s_1 being the end-of-tape marker):

   - the constant symbol 0;
   - the unary function symbol s ('successor' on natural numbers);
   - the binary predicate letter $x < y$ ('number x is strictly less than number y');
   - the binary predicate letter $C(x, y)$ ('at step x, the machine M scans cell tape number y');
   - the unary predicate letter $Q_k(x)$ ('at step x, the machine M is in state q_k');
   - the binary predicate letter $S_k(x, y)$ ('at step x, the cell y contains symbol s_k').

   (a) (5 marks) Write a formula describing the initial configuration of M on the empty word given as input.

   (b) (5 marks) Write a formula saying that, at every step of the computation, M is in a unique state.

   (c) (5 marks) Suppose that the program of M contains the command $q_i s_k \to q_j s_l \mathrm{R}$. Write a formula describing the effect the execution of this command has on the computation.